

Certified Frontiers: Self-Expanding Positivity Certificates with a Verified Coverage Theorem

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ABSTRACT

Formal verification offers two currencies for a computational fact checked far beyond the reach of proof: a theorem (all-or-nothing) or a one-shot certified computation (this instance). We identify and build a third, which we call a *certified frontier*: a verified procedure whose state is a positive-definiteness certificate, whose expansion step extends that certificate by one site under a single scalar gate proved sound once, whose every prefix is a standalone theorem, and whose halting yields an explicit certified counterexample. We implement the object in Lean 4/Mathlib as a certificate-carrying `GramState` with a gated expand operation and an inertia meter that is the same engine read at finer resolution — so certifier and instrument coincide. We then prove a *coverage theorem* for a concrete instantiation on a fixed analytic kernel of screw type: an explicit family of grids on which the kernel Gram matrix is positive definite with an explicit margin, so the self-expander provably runs to completion. We also report how the coverage theorem was obtained: a pre-registered statement was machine-*refuted*, re-posed via a quantitative diagonal-dominance mechanism, and delivered at an explicit finite rung — with the prover discovering the dominance bound exactly tight there and certifying positivity by another route rather than misattribute the method. We argue the certified frontier is a reusable pattern for verified incremental computation — anytime, resumable, with meaningful prefixes and honest halting — of which streaming positive-definiteness monitoring and active-learning-style kernel exploration are natural applications.

1 INTRODUCTION

Many computational facts are checked far beyond the reach of proof: an inequality verified on ever-larger domains, a positivity property confirmed on ever-finer grids. The standard verification currencies are two: prove the theorem (an all-or-nothing $\forall$ -statement), or certify one instance (a one-shot computation, Flyspeck-style). Between them sits a gap that matches the actual situation of such problems, where what one has is an ever-extending frontier of confirmed instances rather than either a proof or a single check.

This paper identifies and builds an object for that gap, which we call a *certified frontier*: a verified procedure whose state is itself a certificate, whose one-step extension is proved sound once and thereafter requires only a local scalar check,

whose every prefix is a standalone theorem, and whose halting is meaningful — it yields an explicit certified counterexample at the horizon where it stops. The point is not speed (though an executable variant gives that); it is that the frontier converts “verified numerically up to X ,” which is ordinarily a sociological claim about floating-point runs, into “kernel-certified up to X by a procedure whose extension rule is a theorem,” a resumable and composable artifact.

We instantiate the object on a concrete, fixed analytic kernel of screw type: from an even function Ψ with $\Psi(0) = 0$ one forms the Krein Gram kernel $G(t, u) = \Psi(t) + \Psi(u) - \Psi(t - u)$, whose positive semidefiniteness on finite grids is the property the frontier certifies. The particular Ψ we use is defined explicitly in Section 3 and is imported, with its certified numerical bounds, from a companion development; its wider significance is developed there and is not needed here. What matters for this paper is that Ψ is a fixed, explicitly computable analytic kernel on which the frontier’s coverage question — how far the certified self-expander provably runs — is nontrivial.

The paper’s second contribution is the arc by which its coverage theorem was obtained, because it exhibits the frontier methodology honestly. A pre-registered coverage statement was *refuted by the prover* on two independent grounds; re-posed through a different mechanism; delivered at a finite rung; and, at that rung, the prover discovered that the intended mechanism is *exactly* tight and proved the result another way rather than misattribute it. We report all of this as first-class, mirroring the project’s discipline that walls are information.

Contributions.

- **The certified frontier object**, in Lean 4: a certificate-carrying state `[R_A2:GramState]` with a positive margin `[R_A2:margin_le]`; a gated one-site expansion whose soundness is the Schur-complement criterion `[R_A1:schur_pos_isPDq]` packaged as `[R_A3_A4:expand_pd]`; an honest-halt theorem producing an explicit counterexample witness `[R_A3_A4:halt_not_pd]`; and an inertia meter with positive-definiteness as its zero-negative-count corollary `[R_A5:isPDq_iff_nMi]` so certifier and instrument are one engine.
- **A verified coverage theorem for the instantiated kernel**: an explicit grid family on which the Gram matrix is positive definite with explicit margin

[R5Final:coverage_band_final], and its frontier form built literally by singleton and repeated expand [R5Final:frontier_covers_band_final]. A general quantitative diagonal-dominance Rayleigh-floor lemma [R5Prime:gershgorin_margin] underlies it.

- **The refute–repair record** as a methodology result: the machine-checked obstructions [R5:PsiArch_not_convex] and [R5:coverage_prime_free] (the original statement is false), the re-posed and delivered rungs, and the tightness discovery [R5Final:three_grid_last_row_gershgorin_zero].
- **The genre argument:** the certified frontier as a reusable pattern for Π_1 claims, with an executable-reflection seed [R_B1:checkPDq_sound] and a map of instantiations.

The development accompanies the paper as an anonymized artifact; every result carries a [file:declaration] tag into its index. The instantiation imports the screw-kernel definition and validation corpus from the companion paper; those objects are cited, not reproduced.

2 THE CERTIFIED FRONTIER OBJECT

2.1 State as certificate

The core datatype bundles a symmetric matrix with its own positive-definiteness proof and a certified positive margin:

```
structure GramState where
  dim :
  M : Matrix (Fin dim) (Fin dim)
  hpd : IsPDq M
  ...
```

so that *possessing a GramState is possessing a proof*. The margin field satisfies [R_A2:margin_le]: $\text{margin } G \leq \lambda_{\min}(G.M)$, a certified floor. The one-point state is [R_A2:singleton], a 1×1 certificate from a positive scalar.

2.2 Gated self-expansion

Extension adds one site. Given a border (b, d) and the Schur complement $s = d - b^T M^{-1} b$, the criterion [R_A1:schur_pos_isPDq] (with converse [R_A1:border_pd_schur_pos]) says the bordered matrix is positive definite iff M is and $s > 0$. The expansion [R_A3_A4:expand_pd] consumes a proof of $s > 0$ and returns the larger certified state; the faithfulness of the extension (the old block is preserved, the border is (b, d)) is proved alongside. Self-expansion is thus total on the gate and never produces an unproven state: the certificate propagates automatically.

2.3 Honest halt

If the gate fails ($s \leq 0$) the extension is *not* positive definite, and the object says so constructively: [R_A3_A4:halt_not_pd]

exhibits the explicit vector $(-M^{-1}b, 1)$ on which the bordered quadratic form is $s \leq 0$. A stall is therefore a certified counterexample at that horizon, not an inconclusive result — the property that makes the frontier a semi-decision procedure with meaningful failure, and `expand_gate_iff_pd` records the exact equivalence between the gate and border-positivity.

2.4 The inertia meter: certifier equals instrument

The same spectral data that certifies also measures. Defining eigenvalue counts (n_+, n_0, n_-) relative to a threshold, the certifying readout is the special case $n_- = n_0 = 0$: [R_A5:isPDq_iff_nMinus_nZero_eq_zero] proves positive definiteness is exactly the vanishing of the nonpositive counts, and the readout’s invariance under congruence is proved for the positive-definite case. Read coarsely the engine certifies; read finely it is a width/rank instrument. The instrument never enters the objective — it is the certifier observed at resolution.

2.5 Validation against the true kernel

The machinery reconstructs a hand-proved true-kernel window by stepping: [R_A6:gridGate1], [R_A6:gridGate2] discharge the two Schur gates for the grid $(0.2, 0.4, 0.6)$, and singleton followed by two expands rebuilds the audited three-point Gram of the companion paper’s screw kernel, matrix-equal. The self-expander is thus validated to agree with a bespoke certificate on the genuine object.

3 INSTANTIATION: A FIXED SCREW-TYPE KERNEL

The frontier is instantiated on $G(t, u) = \Psi(t) + \Psi(u) - \Psi(t - u)$ for a fixed even analytic Ψ with $\Psi(0) = 0$, imported with its certified numerical bounds from a companion development. The kernel has a smooth “base” part and, beyond a threshold $t_0 > 0$, a piecewise-linear correction; below that threshold (the interval on which our coverage theorem operates) the base part alone governs, and it is explicitly computable. We treat Ψ purely as a fixed analytic kernel; its provenance and wider significance are developed in the companion paper [1] and are not used here. The frontier’s coverage question is how far the certified self-expander provably runs on this kernel.

4 THE COVERAGE THEOREM, AND HOW IT WAS OBTAINED

This section reports an arc, because the arc is a result.

4.1 Attempt one, refuted

The pre-registered statement asked for a mesh δ and margin μ such that *every* grid in the base-only zone with spacing $\geq \delta$ yields a positive-definite Gram with floor μ , via an assumed convexity of the archimedean profile. The prover refuted it twice over. First, [R5:PsiArch_not_convex]: the profile is *not* convex even on $[0, 0.4]$ — from the audited bounds $\Psi(0.2) > 0.0543$ while $\Psi(0.4) < 0.0413$, so it decreases where a convex-through-zero function cannot. Second, [R5:coverage_prime_free]: the all-grid statement is outright false, since it admits the singleton grid $\{0\}$ whose Gram is the zero matrix ($G(0, 0) = 2\Psi(0) = 0$), so no positive μ can exist. Per the discipline this was reported, not patched: the convexity route was abandoned and the statement re-posed.

4.2 Attempt two: diagonal dominance, honest mesh

The repair replaces convexity with a quantitative diagonal-dominance floor, proved in general: [R5Prime:gershgorin_margin] shows that a symmetric matrix with $\mu + \sum_{j \neq i} |M_{ij}| \leq M_{ii}$ in every row has $\mu \sum x_i^2 \leq x^T M x$. This is a quantitative Rayleigh-floor strengthening of the diagonal-dominance direction of Gershgorin’s theorem (Mathlib provides the circle theorem and strict-dominance nonsingularity; the explicit-margin PSD form is what coverage needs). Instantiated on a non-degenerate band $[1/5, 3/5]$ with the origin excluded, it yields a certified two-site window [R5Prime:coverage_band] at separation $\delta = 2/5$, margin $\mu = 3/100$, with the mesh condition stated explicitly rather than hidden.

4.3 Attempt three: the three-site rung and a tightness discovery

Extending to three sites forces the grid $(1/5, 2/5, 3/5)$, and here the prover found something sharp: [R5Final:three_grid_last_row_diagonal_dominance_rule] proves the last row’s diagonal-dominance margin is *exactly* zero — $G(t_2, t_2) = G(t_2, t_0) + G(t_2, t_1)$ on the nose — so the general diagonal-dominance lemma cannot supply a *positive* three-site floor. The matrix is nonetheless positive definite (diagonal dominance is sufficient, not necessary), and [R5Final:coverage_band_final] certifies it with $\delta = 1/5$, $\mu = 1/200$ by invoking the companion paper’s audited eigenvalue margin — not by claiming a Gershgorin floor the prover had just shown to be absent. The frontier form [R5Final:frontier_covers_pansion_rule] builds the three-site state by singleton and two expands. The tightness discovery is itself the honest boundary: the diagonal-dominance mechanism is exactly exhausted at this mesh.

4.4 The honest boundary

Coverage is established on coarse, separated grids; the fine-grid ($\delta \rightarrow 0$) regime is neither covered nor expected to be by a diagonal-dominance argument, and this is the natural boundary rather than a failure — coarse grids are the correct object for the mechanism. The $\forall$ -over-all-scales limit of the coverage statement — positivity on arbitrarily fine grids — is the hard direction; the frontier proves the first rungs and names precisely where the diagonal-dominance mechanism stops.

5 EXECUTABLE REFLECTION (SEED)

The frontier above is proof-term-based: each window is a proof. An executable variant makes windows kernel evaluations. We deliver the first component: a Boolean rational Sylvester checker [R_B1:checkPDq] with soundness [R_B1:checkPDq_sound] (checkPDq $M = \text{true}$ implies the real cast of M is positive definite), reducing under decide over $\mathbb{Q}$ with no native_decide. The interval-entry bridge and the evaluation-based re-proof of the true-kernel window are specified and left as the engine’s completion; the seed establishes the soundness pattern.

6 THE CERTIFIED FRONTIER AS A GENRE

The object generalizes past this instance. Abstractly a certified frontier is an invariant-carrying incremental certificate: state that is a proof, a soundness theorem for one extension step, meaningful prefixes, and honest halting. Its natural targets are Π_1 claims presented as exhaustions — Li-coefficient positivity, screw-moment Hankel-determinant positivity, zero counting by the argument principle, and, in applied form, streaming certified positive-definiteness of kernel/covariance matrices. In each the same three pieces recur: an extension rule (proved once), a coverage/margin theorem (new mathematics, the effective-constants content), and the $\forall$ -limit (the conjecture).

We note the closest neighbors honestly. Rigorous positive-definiteness verification via one Cholesky with rounding analysis is classical (Rump); verified Cholesky exists in ACL2 (Kwan); interval arithmetic exists in Lean; incremental pivoted Cholesky with free-energy pivot selection is current numerics (Schaub–Zaspel), an uncertified sibling of our expansion rule; and a program whose halting encodes an open statement is known (Yedidia–Aaronson) but with opaque prefixes. What the certified frontier adds is the conjunction: kernel-checked, incremental, certificate-as-state, meaningful prefixes, and a coverage theorem as a first-class target.

7 DISCUSSION AND FUTURE DIRECTIONS

A stateful frontier tactic. The object packages naturally as a resumable, anytime decision procedure for exhaustion-indexed positivity goals — a tactic that answers “covered up to horizon a ?” by resuming a persistent certified state. Existing decision procedures are stateless; this is a distinct genre.

Semi-decision with productive prefixes. For Π_1 statements admitting a frontier decomposition, the object is an anytime semi-decider: certified prefixes throughout, certified refutation on halt. This is the precise sense in which it is “prover-like” without being a prover — its authority derives entirely from the kernel that checks its soundness theorem.

Frontier-scheduled search (speculative). One could schedule proof search by the conditional-variance heuristic that schedules the frontier’s expansion; but the incorruptibility that makes the frontier’s state a certificate does not transfer to a value model over open goals, so this belongs in the proposer/oracle slot of a neural-proposes/kernel-checks split, where Bayesian premise selection is already studied. We flag it as a direction, not a result. A further obstruction is intrinsic to the object: a selector reading the objective’s own margin is blind precisely at the tight configurations where positivity is hardest to certify — the exactly-zero diagonal-dominance margin of Section 4 ([R5Final:three_grid_last_row_gershgorin_zero]) is such a site, certifiable only off the local gate — so an incorruptible self-steering frontier, if one exists, would have to schedule on a loop-side meter with a measured floor rather than on the certificate it extends. Whether such a scheduler provably beats random ordering, or provably cannot, is the natural next question.

Completing the instance. The immediate concrete work: finish the executable interval engine (the bridge and evaluation-based windows), and push the coverage mesh where a mechanism beyond diagonal dominance permits — noting that the fine-grid regime is the genuinely hard direction.

8 RELATED WORK

Screw-type kernels and their positivity theory (Krein); the companion development in which this paper’s kernel and its certified bounds originate [1]; verified positive-definiteness and Cholesky (Rump; Kwan; Mathlib’s Gershgorin and spectral theory); rigorous interval numerics in proof assistants; incremental pivoted Cholesky with free-energy pivot selection in Gaussian-process numerics (Schaub–Zaspel); and programs whose halting encodes an open statement (Yedidia–Aaronson). The certified-frontier genre is, to our knowledge,

not assembled elsewhere; each component is individually occupied.

ACKNOWLEDGMENTS

The Lean 4 proofs were produced by an automated theorem prover from human-written specifications and human-audited statement-by-statement; the prose is the author’s. Per-batch provenance is in the artifact.

REFERENCES

- [1] Anonymous. 2026. Certified Positivity: Formally Verified Horizon Thresholds for Truncated Weil Quadratic Forms. Companion paper.

A ARTIFACT AND DECLARATION INDEX

The anonymized artifact contains the full Lean development (shared with the companion paper; this paper’s results are the `R_*` and `R5*` modules). Every `[file:declaration]` tag in the text resolves in the artifact README’s index. The R-tier modules are `sorry/admit/ native_decide-free`; headline theorems use only `propext`, `Classical.choice`, `Quot.sound`.